QuantPulse

AI-Powered Short-Term Stock Trend Forecasting

Product overview: features, technology, architecture, operating guide, and measured forecasting results across a 10-ticker universe.

Coverage: AAPL, AMZN, FDX, GOOGL, IBM, META, MSFT, NVDA, ORCL, TSLA | Generated from live training artifacts

1. Product overview

QuantPulse is a forecasting platform for retail and professional traders. It combines market price history (OHLCV) with financial-news sentiment to produce 5-day short-term closing-price forecasts with 95% confidence bands, 30-day longer-horizon projections, per-ticker directional trend signals (UP/DOWN), and a model-comparison dashboard that benchmarks the hybrid deep model against five classical baselines.

- 5-day forecasts with confidence bands + 30-day horizon per ticker.
- Directional trend accuracy (up/down) evaluated on a held-out test period.
- Sentiment overlay: FinBERT-tone news tone aligned to trading days.
- Ablation dashboard: LSTM vs LSTM+Attention vs +Sentiment vs Full Hybrid with federated averaging.
- FastAPI backend + React/Tailwind dashboard; one-click run.bat and Docker support.

2. Key features

- Configurable ticker universe (default: TSLA, AAPL, AMZN, MSFT, GOOGL, NVDA, META, FDX, IBM, ORCL).
- Live OHLCV ingestion via Yahoo Finance with bundled-CSV and labeled-synthetic fallbacks.
- NewsAPI ingestion with local caching; FinBERT-tone sentiment scoring (inference only).
- Simulated federated training (per-ticker clients + FedAvg) runnable on a laptop CPU.
- Input validation and graceful errors across all API endpoints.

3. Technology stack

Layer	Technology
Backend API	Python 3.11, FastAPI, Uvicorn (127.0.0.1:8000)
Deep models	TensorFlow/Keras: LSTM-64 + 4-head attention
Baselines	statsmodels (ARIMA/SARIMAX), Prophet, scikit-learn (RF/SVR)
Sentiment	NewsAPI + FinBERT-tone (transformers, PyTorch CPU)
Frontend	React 18, Vite, Tailwind CSS, Recharts
Packaging	run.bat, Dockerfile, docker-compose.yml

4. Architecture summary

OHLCV (20-day Min-Max windows) and daily FinBERT sentiment form the model input. Five classical baselines are scored on an 80/20 chronological split. The deep track trains LSTM, LSTM+Attention, and LSTM+Attention+Sentiment tiers per ticker; a federated stage then averages per-ticker client weights with FedAvg (2 rounds, in-process simulation). Recursive multi-step decoding produces the 5- and 30-day forecasts; residual-based bands give the 95% intervals. The FastAPI layer serves forecasts, metrics, sentiment, and history; the React SPA renders charts, the comparison dashboard, and the sentiment panel.

5. How to run

- Automatic: double-click run.bat, then open http://localhost:8000.
- Docker: docker compose up --build, then open http://localhost:8000.
- Manual: create venv (3.11), pip install -r backend/requirements.txt, python -m backend.train, then uvicorn

backend.app.main:app --host 127.0.0.1 --port 8000.

- Full instructions and troubleshooting: HOW_TO_RUN.md.

6. Results and metrics (measured)

All values below were produced by executing backend/train.py (80/20 chronological split; deep tiers evaluated on windowed test sets). Data mode(s) in this build: live. Sentiment scorer: finbert; news source: synthetic-fallback.

6.1 Ablation - deep model tiers (MAE / directional accuracy)

Ticker	LSTM	+Attn	+Sent	Hybrid(Fed)
AAPL	51.72/90.6%	14.64/90.6%	14.96/90.6%	30.60/90.6%
AMZN	17.77/79.2%	11.63/79.2%	30.23/79.2%	25.98/79.2%
FDX	98.79/81.2%	58.25/81.2%	22.44/81.2%	38.26/81.2%
GOOGL	89.54/86.5%	14.84/89.6%	14.70/21.9%	60.10/86.5%
IBM	23.93/56.2%	21.45/55.2%	24.91/59.4%	23.35/55.2%
META	48.10/52.1%	33.99/57.3%	29.80/52.1%	34.85/52.1%
MSFT	43.03/57.3%	31.03/52.1%	27.30/50.0%	35.19/50.0%
NVDA	45.18/76.0%	8.46/81.2%	7.51/82.3%	30.53/76.0%
ORCL	23.78/55.2%	24.97/43.8%	76.50/44.8%	23.46/43.8%
TSLA	35.70/53.1%	28.46/53.1%	26.93/52.1%	37.00/50.0%

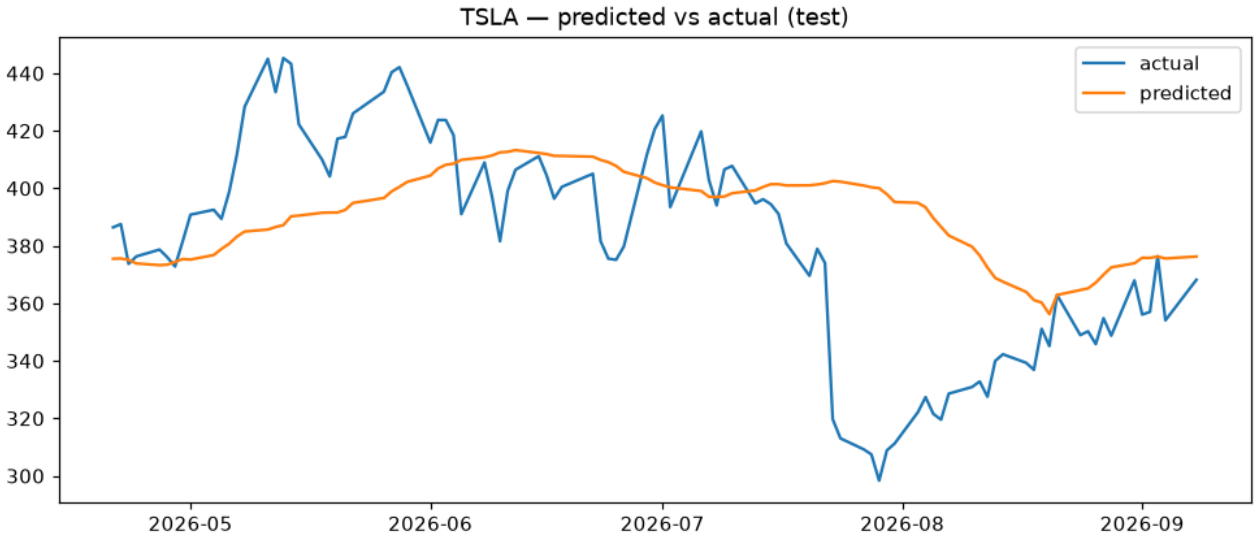
6.2 Classical baselines (MAE / directional accuracy)

Ticker	ARIMA	SARIMAX	Prophet	RF	SVR
AAPL	44.66/45.0%	41.29/45.0%	59.90/45.0%	23.52/50.0%	50.60/48.0%
AMZN	12.02/59.0%	12.89/49.0%	54.11/55.0%	15.28/57.0%	31.56/55.0%
FDX	20.63/57.0%	14.29/46.0%	44.38/48.0%	13.86/55.0%	41.90/53.0%
GOOGL	23.12/60.0%	24.64/51.0%	62.60/54.0%	22.68/58.0%	59.53/54.0%
IBM	21.91/50.0%	23.81/50.0%	57.58/48.0%	6.30/53.0%	12.13/54.0%
META	64.71/54.0%	82.37/51.0%	33.31/53.0%	12.74/45.0%	15.81/49.0%
MSFT	42.05/56.0%	56.45/48.0%	143.99/47.0%	8.03/50.0%	12.91/49.0%
NVDA	16.85/54.0%	11.37/60.0%	32.29/52.0%	11.69/54.0%	29.53/52.0%
ORCL	24.81/49.0%	26.04/49.0%	80.12/53.0%	7.24/50.0%	11.62/45.0%
TSLA	36.02/54.0%	36.15/52.0%	55.08/50.0%	10.48/51.0%	12.30/52.0%

6.3 Tesla deep dive

Last close $368.16; 5-day outlook: DOWN. Hybrid test MAE 37.00, R2 -0.473, directional accuracy 50.0%. Training/validation loss curves and predicted-vs-actual charts are stored in artifacts/plots/ and embedded in the dashboard screenshots.

Tesla predicted vs actual (test period)



Tesla LSTM+Attention+Sentiment loss curves

